FARAAZ MOHAMMED

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Full Australian working rights (Temporary Graduate visa, subclass 485) — no sponsorship required • Available immediately

PROFILE

Master of Artificial Intelligence graduate (RMIT, 2026) with a B.Tech in IT; full-stack developer intern at Vayaa after an AI/ML engineering internship delivering geospatial ML pipelines at national scale. Co-author of a peer-reviewed computer-vision paper. Works end to end — requirements, code, tests, review — across Python, Rust, Android, web and AWS; built two Android apps and an MCP-based LLM tool in 2026. Seeking graduate AI/ML or software engineering roles across Australia.

TECHNICAL SKILLS

Languages: Python (primary), Kotlin, JavaScript, TypeScript, Rust, SQL, Prolog / ASP (Clingo), JCL, Bash

AI / ML: scikit-learn, XGBoost, OpenCV (image classification, edge detection, Haar cascades), evaluation design (precision/recall, IoU, sensitivity analysis, ground-truth construction), LLM fine-tuning (Unsloth, QLoRA, GGUF), Ollama, Answer Set Programming, prompt engineering, retrieval-grounded generation

Geospatial: GeoPandas, Shapely, QGIS, PostGIS, Overture Maps, OpenStreetMap, GeoPackage / GeoJSON / Parquet

Software: Android (Kotlin, AlarmManager, BroadcastReceiver, notification channels, Gradle KTS, JUnit), web APIs (FastAPI, Rust/Axum), PostgreSQL (SQLAlchemy), React, PWA, Node.js, Next.js, Prisma, REST/JSON

AI-assisted engineering: MCP servers, Claude Code (skills, hooks, subagents), Cursor, Anthropic API (prompt caching, routing)

Tooling & platforms: Git / GitHub, Docker, AWS (CloudFormation, Fargate, RDS, CloudFront, SNS), Kestra, Colab, Vercel, Linux

EXPERIENCE

Full-Stack Developer Intern — Vayaa
Sep 2026 – present

Melbourne, VIC · hybrid · company website: React + TypeScript, Python (FastAPI) + PostgreSQL, deployed on AWS

- Building the company's website from a mentor's brief; turned each plain-language request into numbered functional and non-functional requirements and kept a decision log (choice, alternatives, cost) for every design call.
- Built the contact-enquiry API in Rust (server-side validation, clean types), then led its port to Python (FastAPI) on the mentor's call, gating the cutover on 50 parity tests passing against both backends.
- Took it live on AWS with CloudFormation (CloudFront → ECS Fargate → RDS PostgreSQL) in five attempts, fixing Free-plan limits, a CloudFront account block and RDS TLS (verify-full); SNS emails each enquiry.

Tools: Python, FastAPI, SQLAlchemy, PostgreSQL, Docker, AWS, Rust (Axum), React, TypeScript, Claude Code

AI/ML Engineering Intern — Prodegee
Mar 2026 – Jun 2026

Remote, Melbourne · RMIT industry capstone, Team DS4 — building-footprint inference for off-grid electrification planning

- Designed and built the school-geolocation algorithm at the core of the team's pipeline (7.2M Overture building polygons, BFS spatial clustering, composite scoring); ran it on 18,121 schools in Côte d'Ivoire and ported it to Burundi and Chad.
- Replaced a binary distance cutoff with exponential-decay scoring and a weighted confidence score; added per-country sigmoid normalisation so one threshold works in all three countries; authored cross-team handoff contracts.
- Built a ±20% sensitivity harness that identified cluster radius as the dominant parameter (±36–44% swing across 191 villages), driving a 75 → 200 m recalibration that fixed a 90%+ household under-count (0.069 → 0.415).
- Built an IoU evaluation harness for compound polygons and traced all three generators' failures to a ~35× over-scoped upstream cluster rather than a fitting fault, which redirected the next assignment.
- Ran footprint inference across 200 villages in 7 Mauritanian Wilayas and trained an XGBoost multi-output tier regressor; documented that its 0.816 "confidence" measured agreement with the training heuristic, not accuracy.
- Co-built a 40-school satellite-verified ground-truth set and used it to correct a published baseline (6–13% → 37%); published a negative result showing a proximity penalty hurt accuracy at every threshold.
- Led 3 of 10 team assignments and was primary author of 3 of 4 formal deliverables (68 commits, 7 peer reviews); pinned upstream inputs by commit SHA for reproducible re-runs and wrote the semester-2 handoff memo.
- Built a Claude API model router that classifies task complexity with a cheap Haiku call and routes to Haiku / Sonnet / Opus, keeping cost proportional to difficulty.

Tools: Python, GeoPandas, Shapely, scikit-learn, XGBoost, pandas, Kestra, QGIS, GeoPackage, Git, Claude Code (MCP, hooks)

Dispatch Team Member — Amart Furniture
Feb 2025 – Mar 2026

Melbourne, VIC · part-time, on-site, concurrent with full-time study

- Tracked and reconciled stock in the inventory system, processed customer orders and resolved fulfilment discrepancies across 5 Melbourne stores; trained a new team member.

EDUCATION

Master of Artificial Intelligence — RMIT University
Jul 2024 – Jul 2026

Melbourne, VIC · Intelligent Decision Making (ASP/Clingo), Computer Vision, Machine Learning, AI Systems Design, IoT

Bachelor of Technology (Information Technology) — Keshav Memorial Institute of Technology *Dec 2020 – May 2024*
Hyderabad, India (affiliated to JNTUH) · Coursework incl. Machine Learning, Neural Networks & Deep Learning, Data Mining

KEY PROJECTS

Time Flow — ADHD-focused planner (Android, closed source) · JavaScript, Kotlin *Jul 2026 – present*

- Building an Android planner for commercial release that packs goals, recurring targets and reminders into free minutes by an urgency × importance score; 10 tagged releases, the latest shaped by tester feedback.
- Vanilla-JS UI in a Kotlin WebView shell that owns exact alarms, reboot recovery and a foreground service keeping focus timers alive; iOS (Swift) shell in progress.
- Set up CI that builds a shrunk release APK and runs the Kotlin and JavaScript test suites on every PR; tagged releases are signed and checked with apksigner.

  ↳ Closed source (in development for commercial release) · demo on request

MGM Prayer Alerts — Android app for Melbourne Grand Mosque · Kotlin, AlarmManager, JUnit *Aug – Oct 2026*

- Shipped an Android app that fetches the mosque's daily timetable, schedules exact Azaan alarms with per-prayer toggles and custom sounds, refreshes nightly and survives reboot; in use by community members.
- Wrote functional requirements (scope, decision log, acceptance criteria) before coding, then reviewed the code against them: fixed a critical AM/PM storage bug and a refresh alarm that stopped re-arming; unit-tested time parsing.
- When the timetable source began returning bot-check pages, kept alarms working by falling back to a bundled annual timetable behind a new unit-tested parser; added GitHub Actions to test, build and publish releases.

  ↳ GitHub: github.com/faraaz817/MgmPrayerAlerts

Gemma Delegation System — local/cloud LLM hybrid · Node.js, MCP SDK, Ollama, Unsloth/QLoRA, Colab *May 2026*

- Built an MCP server that lets Claude Code or Cursor delegate lightweight tasks to a locally hosted Gemma model, failing open to the frontier model whenever Ollama is unavailable so delegation never blocks the user.
- Closed the loop: accept/reject feedback updates a capability profile and curates a JSONL dataset; a Colab notebook fine-tunes on confirmed examples (4-bit QLoRA, LoRA r=16) and exports GGUF back into Ollama.

  ↳ GitHub: github.com/faraaz817/gemma-delegation-system
- **Bro — interactive AI portfolio chatbot:** recruiter-facing site that answers questions from a profile corpus via server-side retrieval with anti-hallucination constraints; Anthropic API, deployed on Vercel. (Next.js, TypeScript, Prisma/Postgres)

  ↳ Live: job-go-mauve.vercel.app/portfolio
- **Fire Detection & Alerting System (B.Tech capstone, team of 4):** trained a custom OpenCV Haar cascade (Python) for real-time fire detection, chosen over heavier detectors for latency; a threaded pipeline raises an alarm and sends WhatsApp alerts to the nearest fire station.
- **NeuroVisualZ — exam-timetabling AI, RMIT (team of 3):** built the YAML-to-ASP encoder that translates timetabling instances into Clingo logic programs; scoped Levels 1–4 (room allocation, ITC'07 soft constraints, Prolog validator).
- **Agent skills for Claude Code / Cursor:** authored five reusable skills — CO-STAR requirements builder, Socratic tutor, better-way reviewer, skill sync, and a pikepdf CLI that unlocks signature-locked PDFs. github.com/faraaz817/My_Skills
- **IBM Z development toolchain:** connected VS Code to a live z/OS environment via Zowe CLI/Explorer and z/OSMF REST APIs; submitted and verified a JCL job with condition code 0000. (IBM z/OS, Zowe, JCL)

PUBLICATION

K. M. Asudaria, **M. F. Abdul Khadeer**, S. S. Rafai, K. S. Reddy. "A Comparative Study of Edge Detection Techniques to Identify Maize Leaf Diseases using Machine Learning." *Quest Journal of Software Engineering and Simulation*, Vol. 9(12), pp. 43–51, Dec 2023 (ISSN 2321-3795, peer-reviewed). Benchmarked 5 edge-detection operators × 8 classifiers (40 combinations) as a lesion feature-extraction step on a 15,000-image, four-class maize dataset; best result Sobel + MLP at 94.75%. Implemented the Canny arm (OpenCV + scikit-learn, ten classifiers). ↳ questjournals.org/jses · Code: github.com/faraaz817/Machine_Learning

CERTIFICATIONS, AWARDS & LEADERSHIP

- IBM badges on Credly: IBM Z Xplore – Concepts (May 2026) · IBM Z & LinuxONE Community Contributor – Level 1 (Aug 2026).
- Smart India Hackathon delegate, Government of India (2022) · Certificate of Publication, Quest Journals (2023).
- Head, Vachan Speakers Club, KMIT (2022–23) — led public-speaking workshops, debates and communication seminars.
- Volunteer fundraiser, Taher Foundation (2023).

LANGUAGES

English (native / bilingual) • Urdu (native / bilingual) • Hindi (full professional) • Telugu (elementary)

REFEREES

Available on request.